A Project Report on

PPO-Based Semantic and Age-of-Information-Aware Scheduling for Massive IoT Networks

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

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by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: PPO-Based Semantic and Age-of-Information-Aware Scheduling for Massive IoT Networks

1. Abstract:

Massive Internet of Things (IoT) networks consist of a large number of devices that continuously generate information for applications such as smart cities, healthcare monitoring, industrial automation, and intelligent transportation. In such networks, communication opportunities are limited, so not all devices can transmit their information at the same time. Therefore, an important scheduling problem is deciding which IoT device should be given the next transmission opportunity. Two important factors are the semantic importance of the generated information and its Age of Information (AoI). AoI represents how fresh or stale the information available at the receiver is, while semantic importance indicates how useful or important the information is to the application. A scheduling policy that minimizes AoI alone may prioritize information that is very stale but less important, whereas a policy based only on semantic importance may allow important information to become outdated. Therefore, this project focuses on the joint scheduling trade-off between information freshness and semantic importance in a simulated massive IoT environment. The objective is to investigate whether an intelligent scheduling policy can make better transmission decisions by considering both factors under limited communication opportunities.

To address this problem, the proposed project develops a Proximal Policy Optimization (PPO)-based reinforcement learning scheduling framework. The IoT network is modeled as a sequential decision-making environment with 10 virtual IoT devices as the initial simulation configuration. Each device is represented by its Age of Information and a synthetically generated semantic-importance value. At every discrete time slot, only one device can be selected for transmission. The PPO agent observes the current network state, selects one device, receives a reward based on the resulting scheduling decision, and learns through repeated interaction with the environment. The reward is designed to capture the trade-off between information freshness and semantic importance. The learned PPO scheduler is experimentally compared with three baseline strategies: random scheduling, highest-AoI scheduling, and

highest-semantic-importance scheduling. Performance is evaluated using average AoI, peak AoI, semantic utility, and overall scheduling reward. The results obtained from the simulation will be used to determine how effectively the PPO policy balances freshness and semantic importance compared with the baseline scheduling approaches.

2. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11 or Linux
- Programming Language: Python 3.10 or above
- IDE/Development Environment: VS Code / Jupyter Notebook
- Deep Learning / Reinforcement Learning Framework: PyTorch
- Reinforcement Learning Algorithm: Proximal Policy Optimization (PPO)
- RL Environment: Gymnasium
- Numerical Computing: NumPy
- Data Processing: Pandas
- Data Visualization: Matplotlib, Seaborn
- Machine Learning Utilities: Scikit-learn
- Network Simulation: Python-based custom IoT network simulation environment
- Semantic Analysis: Sentence Transformers / pretrained transformer-based semantic similarity model
- Model and Data Storage: CSV, JSON, and PyTorch model files
- Version Control: Git and GitHub
- Hardware Monitoring/Experimentation: Optional CUDA support with an NVIDIA GPU for faster PPO training

Signature of the Supervisor Course Instructor